

paper2

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1 Introduction

To select the optimal hyperparameters for the proposed skip module, we employ a constrained multi-objective optimization procedure on the validation set. Let R_{base} denote the end-to-end recall and FAR_{base} denote the false alarm rate of the baseline pipeline without skipping. For each candidate parameter set $\theta \in \Theta$ obtained by grid search, we evaluate the full pipeline $\text{Read} \rightarrow \text{Skip}(\theta) \rightarrow [\text{DL}]$ and compute three metrics: recall $R(\theta)$, false alarm rate $\text{FAR}(\theta)$, and negative-frame skip ratio $S(\theta)$.

The negative-frame skip ratio is defined as

$$S(\theta) = \frac{\# \text{ correctly skipped negative frames}}{\# \text{ total negative frames}}.$$

To combine multiple objectives in a single score, the metrics should be expressed on comparable scales. Accordingly, we define the normalized false alarm reduction as

$$\Delta \widetilde{\text{FAR}}(\theta) = \max \left(0, \frac{\text{FAR}_{\text{base}} - \text{FAR}(\theta)}{\text{FAR}_{\text{base}}} \right),$$

which measures the relative reduction in false alarm rate with respect to the baseline system. Under the conservative-gating assumption of the proposed pipeline, $\text{FAR}(\theta) \leq \text{FAR}_{\text{base}}$ should hold theoretically; the $\max(0, \cdot)$ form is retained for robustness and implementation safety.

To reflect recall preservation in a simpler and more interpretable way, we define the recall-retention term as

$$\widetilde{R}(\theta) = 1 - \frac{R_{\text{base}} - R(\theta)}{\delta_R},$$

where δ_R is the maximum allowable absolute recall drop. This term equals 1 when the candidate matches the baseline recall and equals 0 when the candidate reaches the lowest acceptable recall level $R_{\text{base}} - \delta_R$.

The optimal parameter set θ^* is selected as

$$\theta^* = \arg \max_{\theta \in \Theta} \left[w_S S(\theta) + w_F \Delta \widetilde{\text{FAR}}(\theta) + w_R \widetilde{R}(\theta) \right] \quad \text{subject to} \quad R(\theta) \geq R_{\text{base}} - \delta_R,$$

where $\delta_R = 0.01$ denotes a 1% absolute recall-drop tolerance and the nonnegative weights satisfy

$$w_S + w_F + w_R = 1.$$

In this work, we set

$$w_S = 0.60, \quad w_F = 0.20, \quad w_R = 0.20,$$

so that skip ratio remains the primary efficiency objective, while false alarm reduction and recall retention are treated as secondary but still explicit preferences. This formulation preserves the safety-first role of recall through the hard constraint, while avoiding the drawback of selecting among feasible candidates using skip ratio alone.

Theoretical justification. The skip module acts as a conservative gate: skipped frames are forced to output “negative,” while passed frames are processed by the same downstream detector as in the baseline system. Therefore, the false alarms of the skip-enabled system form a subset of those of the baseline system, implying $\text{FAR}(\theta) \leq \text{FAR}_{\text{base}}$ under the assumed architecture. The main safety risk is thus recall degradation due to false skips, which motivates using recall as both a hard feasibility condition and a soft ranking term.

Algorithm 1 Skip Module Hyperparameter Selection

Require: parameter space Θ , validation set D_{val} , detector M , recall tolerance δ_R , weights w_S, w_F, w_R

Ensure: optimal parameters θ^*

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1: baseline  $\leftarrow$  evaluate_pipeline( $D_{\text{val}}$ , skip=None,  $M$ )
2: extract  $R_{\text{base}}, \text{FAR}_{\text{base}}$  from baseline
3: best_score  $\leftarrow -\infty$ 
4: best_θ  $\leftarrow$  None
5: for  $\theta \in \Theta$  do
6:   skip  $\leftarrow$  SkipModule( $\theta$ )
7:   results  $\leftarrow$  evaluate_pipeline( $D_{\text{val}}$ , skip,  $M$ )
8:   extract  $R(\theta), \text{FAR}(\theta), S(\theta)$  from results
9:   if  $R(\theta) \geq R_{\text{base}} - \delta_R$  then
10:      $\widetilde{\Delta\text{FAR}}(\theta) \leftarrow \max\left(0, \frac{\text{FAR}_{\text{base}} - \text{FAR}(\theta)}{\text{FAR}_{\text{base}}}\right)$ 
11:      $\widetilde{R}(\theta) \leftarrow 1 - \frac{R_{\text{base}} - R(\theta)}{\delta_R}$ 
12:     score  $\leftarrow w_S S(\theta) + w_F \widetilde{\Delta\text{FAR}}(\theta) + w_R \widetilde{R}(\theta)$ 
13:     if score > best_score then
14:       best_score  $\leftarrow$  score
15:       best_θ  $\leftarrow$  θ
16:     end if
17:   end if
18: end for
19: return best_θ

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2 Results

Table 1 shows an example of the validation-time ranking. The selected configuration θ_1^* satisfies the recall constraint and achieves the highest composite score by jointly balancing skip ratio, false alarm reduction, and recall retention.

Table 1: Example hyperparameter selection results

Param Set	Recall Drop	FAR Red.	Skip Ratio	Score	Selected
θ_1^*	0.4%	42%	87.3%	0.728	✓
θ_2	0.8%	45%	84.1%	0.635	×
θ_3	1.2%	51%	82.5%	—	×

This formulation is systematic, interpretable, and aligned with the intended role of the skip module in real-time fire/smoke detection: preserve recall first, then prefer candidates that skip more negative frames while still improving operational false alarm behavior.